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Empirical Reevaluation of Computational Limits in Exact NMR Spectral Simulation: 16 Spin-½ Nuclei on Standard Hardware

Dmitry A. Cheshkov[1], Dmitry O. Sinitsyn[2]

[1]State Scientific Research Institute of Chemistry and Technology of Organoelement Compounds, 38 Shosse Entuziastov, 105118 Moscow, Russia

[2]Russian Center of Neurology and Neurosciences, 80 Volokolamskoe shosse, 125367 Moscow, Russia

E-mail: dcheshkov@gmail.com[1], d_sinitsyn@mail.ru[2]

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To assess the computational limits of a modern personal computer for the exact calculation of NMR spectra of large coupled spin systems, we performed a series of benchmark simulations. Contrary to the common belief that personal computers can only simulate spectra for spin systems of up to 12 spins, we have successfully calculated full spectra for systems of 13, 14, 15, and even 16 spin-½ nuclei (via full Hamiltonian diagonalization). The results demonstrate that a full spectral simulation for a 16 spin-½ system is feasible on a standard high-end workstation (Intel Core i9) in approximately 11.6 hours. The algorithmic complexity and memory requirements are analyzed, confirming that matrix diagonalization remains the primary bottleneck of the NMR spectra calculation problem.

The exact simulation of Nuclear Magnetic Resonance (NMR) spectra for large spin systems is a computationally intensive task, limited by the exponential growth of the Hilbert space. While approximate methods exist, exact diagonalization is often required for strongly coupled systems. To determine the practical limits of solving this simulation problem on modern desktop hardware, we conducted benchmark tests (see Appendix) using the ANATOLIA v1.2 [1, 2] software calculation kernel.

The simulations covered systems ranging from 12 to 16 spin-½ nuclei. The benchmarks were performed on a workstation equipped with an Intel Core i9-10920X CPU (3.50GHz), running Linux 6.17.9. The code was compiled using gcc v15.2.1 and linked against the GSL library v2.7. The summary of computational performance is presented in Table 1.

Table 1. Benchmark results for simulating spin-½ systems of size N = 12–16.

N Spins	RAM	Find Eigensystem	CalcFreqIntens	CalcFullSpectrum	MaxBlockSize/ DiagonalizationTime	Total simulation time
12	143M	3.5	1.1	1.8	924 / 1.3	6.5
13	516M	31.3	10	5.5	1716 / 21	47
14	1.96G	556.6	92	16	3432 / 327	665
15	7.55G	5772	712	20	6435 / 3588	6506
16	28.6G	36434	5436	42.5	12870 / 12010	41920

Time is given in seconds.

Theoretical analysis and benchmark data confirm that finding eigenvectors and eigenvalues (diagonalization) is the most computationally demanding operation. The algorithmic complexity for simulating a system of N spin-½ nuclei scales as $O(\frac{8^N}{\sqrt{N^3}})$.

Memory consumption (RAM) follows the expression $O(\frac{4^N}{N})$, dictated by the storage of the Hamiltonian and eigenvector matrices. Specifically, the GSL library must allocate memory for the source matrix, the matrix of eigenvectors, and the array of eigenvalues. Consequently, the total amount of memory used can be estimated with good accuracy (in bytes) as $52 \times C_{2N}^{N}$, where the coefficient of 52 reflects the storage of double-precision matrices, integer arrays, and associated allocation overheads. In particular, the memory needed to store the source matrix and the eigenvector matrix can be cut at least twofold.

The successful simulation of a 16 spin-½ system in under 12 hours shows that exact spectral calculation for large molecules is becoming accessible without the need for supercomputing clusters.

References

1. D.A. Cheshkov, K.F. Sheberstov, D.O. Sinitsyn, V.A. Chertkov, "ANATOLIA: NMR software for spectral analysis of total lineshape", *Magn. Reson. Chem.*, **2018**, *56*, 449-457, DOI: 10.1002/mrc.4689.

2. ANATOLIA v1.2 can be downloaded from https://github.com/dcheshkov/ANATOLIA/tree/master/ANATOLIA_V1.2

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Appendix: Execution Logs

Detailed timing logs for each simulation run are provided below.

Compilation string: c++ anatolia12_timer.cpp libgsl.a libgslcblas.a

```
[12Spins, 143M RAM]
[ 0.000s] Program started
[ 0.251s] Hamiltonian constructor completed: 237.1 ms
[ 0.260s] Hamiltonian::Build() completed: 7.3 ms
[ 0.260s] Hamiltonian::FindEigensystem() started...
  Block '1' (size 1x1): 0.0 ms
  Block '2' (size 12x12): 0.0 ms
  Block '3' (size 66x66): 0.6 ms
  Block '4' (size 220x220): 16.9 ms
  Block '5' (size 495x495): 252.1 ms
  Block '6' (size 792x792): 784.2 ms
  Block '7' (size 924x924): 1.359 s
  Block '8' (size 792x792): 821.1 ms
  Block '9' (size 495x495): 256.1 ms
  Block '10' (size 220x220): 17.1 ms
  Block '11' (size 66x66): 0.6 ms
  Block '12' (size 12x12): 0.0 ms
  Block '13' (size 1x1): 0.0 ms
[ 3.767s] Hamiltonian::FindEigensystem() completed: 3.507 s
[ 9.351s] Hamiltonian::CalcFreqIntens() completed: 5.583 s
Number of Transitions: 27490
Number of Points to calculate: 131072
[29.021s] Spectrum::CalcFullSpectrum() completed: 9.955 s
==================================================
Total simulation time: 29.021 seconds

[13 Spins, 516M RAM]
[ 0.000s] Program started
[ 0.807s] Hamiltonian constructor completed: 793.4 ms
[ 0.829s] Hamiltonian::Build() completed: 21.0 ms
[ 0.830s] Hamiltonian::FindEigensystem() started...
  Block '1' (size 1x1): 0.0 ms
  Block '2' (size 13x13): 0.0 ms
  Block '3' (size 78x78): 0.9 ms
  Block '4' (size 286x286): 36.9 ms
  Block '5' (size 715x715): 826.6 ms
  Block '6' (size 1287x1287): 4.367 s
  Block '7' (size 1716x1716): 10.728 s
  Block '8' (size 1716x1716): 10.832 s
  Block '9' (size 1287x1287): 4.424 s
  Block '10' (size 715x715): 799.3 ms
  Block '11' (size 286x286): 36.4 ms
  Block '12' (size 78x78): 1.0 ms
  Block '13' (size 13x13): 0.0 ms
  Block '14' (size 1x1): 0.0 ms
[32.882s] Hamiltonian::FindEigensystem() completed: 32.052 s
[76.215s] Hamiltonian::CalcFreqIntens() completed: 43.333 s
Number of Transitions: 81505
Number of Points to calculate: 131072
[105.859s] Spectrum::CalcFullSpectrum() completed: 29.644 s
==================================================
Total simulation time: 105.861 seconds
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```
[14 Spins, 1958M RAM]
[ 0.000s] Program started
[ 3.051s] Hamiltonian constructor completed: 3.036 s
[ 3.120s] Hamiltonian::Build() completed: 68.1 ms
[ 3.120s] Hamiltonian::FindEigensystem() started...
  Block '1' (size 1x1): 0.0 ms
  Block '2' (size 14x14): 0.0 ms
  Block '3' (size 91x91): 1.4 ms
  Block '4' (size 364x364): 79.3 ms
  Block '5' (size 1001x1001): 2.081 s
  Block '6' (size 2002x2002): 26.724 s
  Block '7' (size 3003x3003): 171.785 s
  Block '8' (size 3432x3432): 157.218 s
  Block '9' (size 3003x3003): 168.622 s
  Block '10' (size 2002x2002): 25.533 s
  Block '11' (size 1001x1001): 2.060 s
  Block '12' (size 364x364): 76.6 ms
  Block '13' (size 91x91): 1.4 ms
  Block '14' (size 14x14): 0.0 ms
  Block '15' (size 1x1): 0.0 ms
[557.301s] Hamiltonian::FindEigensystem() completed: 554.181 s
[892.446s] Hamiltonian::CalcFreqIntens() completed: 335.145 s
Number of Transitions: 204195
Number of Points to calculate: 131072
[966.584s] Spectrum::CalcFullSpectrum() completed: 74.138 s
============================================
Total simulation time: 966.586 seconds
```

```
[15 Spins, 7551M RAM]
[ 0.000s] Program started
[11.885s] Hamiltonian constructor completed: 11.868 s
[12.130s] Hamiltonian::Build() completed: 243.9 ms
[12.130s] Hamiltonian::FindEigensystem() started...
  Block '1' (size 1x1): 0.0 ms
  Block '2' (size 15x15): 0.0 ms
  Block '3' (size 105x105): 2.1 ms
  Block '4' (size 455x455): 213.2 ms
  Block '5' (size 1365x1365): 5.450 s
  Block '6' (size 3003x3003): 164.902 s
  Block '7' (size 5005x5005): 934.623 s
  Block '8' (size 6435x6435): 1762.567 s
  Block '9' (size 6435x6435): 1788.909 s
  Block '10' (size 5005x5005): 928.478 s
  Block '11' (size 3003x3003): 162.992 s
  Block '12' (size 1365x1365): 5.198 s
  Block '13' (size 455x455): 186.3 ms
  Block '14' (size 105x105): 2.1 ms
  Block '15' (size 15x15): 0.0 ms
  Block '16' (size 1x1): 0.0 ms
[5765.651s] Hamiltonian::FindEigensystem() completed: 5753.521 s
[8285.272s] Hamiltonian::CalcFreqIntens() completed: 2519.621 s
Number of Transitions: 258114
Number of Points to calculate: 131072
[8378.775s] Spectrum::CalcFullSpectrum() completed: 93.503 s
============================================
Total simulation time: 8378.776 seconds
```

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[Compliation with optimization flags]
Compilation string: c++ -O3 -march=native -DNDEBUG -ffast-math -flto=auto -funroll-loops -fomit-frame-pointer anatolia12_timer.cpp libgsl.a libgslcblas.a

```
[12Spins, 143M RAM]
[ 0.000s] Program started
[ 0.059s] Hamiltonian constructor completed: 46.5 ms
[ 0.065s] Hamiltonian::Build() completed: 5.4 ms
[ 0.065s] Hamiltonian::FindEigensystem() started...
  Block '1' (size 1x1): 0.0 ms
  Block '2' (size 12x12): 0.0 ms
  Block '3' (size 66x66): 0.6 ms
  Block '4' (size 220x220): 17.4 ms
  Block '5' (size 495x495): 265.5 ms
  Block '6' (size 792x792): 851.7 ms
  Block '7' (size 924x924): 1.327 s
  Block '8' (size 792x792): 801.3 ms
  Block '9' (size 495x495): 246.4 ms
  Block '10' (size 220x220): 17.0 ms
  Block '11' (size 66x66): 0.6 ms
  Block '12' (size 12x12): 0.0 ms
  Block '13' (size 1x1): 0.0 ms
[ 3.593s] Hamiltonian::FindEigensystem() completed: 3.528 s
[ 4.691s] Hamiltonian::CalcFreqIntens() completed: 1.098 s
Number of Transitions: 26058
Number of Points to calculate: 131072
[ 6.461s] Spectrum::CalcFullSpectrum() completed: 1.769 s
==================================================
Total simulation time: 6.462 seconds

[13 Spins, 516M RAM]
[ 0.000s] Program started
[ 0.183s] Hamiltonian constructor completed: 171.2 ms
[ 0.197s] Hamiltonian::Build() completed: 13.3 ms
[ 0.197s] Hamiltonian::FindEigensystem() started...
  Block '1' (size 1x1): 0.0 ms
  Block '2' (size 13x13): 0.0 ms
  Block '3' (size 78x78): 0.9 ms
  Block '4' (size 286x286): 37.1 ms
  Block '5' (size 715x715): 746.9 ms
  Block '6' (size 1287x1287): 4.454 s
  Block '7' (size 1716x1716): 10.177 s
  Block '8' (size 1716x1716): 10.717 s
  Block '9' (size 1287x1287): 4.352 s
  Block '10' (size 715x715): 795.2 ms
  Block '11' (size 286x286): 36.8 ms
  Block '12' (size 78x78): 0.9 ms
  Block '13' (size 13x13): 0.0 ms
  Block '14' (size 1x1): 0.0 ms
[31.515s] Hamiltonian::FindEigensystem() completed: 31.318 s
[41.497s] Hamiltonian::CalcFreqIntens() completed: 9.983 s
Number of Transitions: 81505
Number of Points to calculate: 131072
[46.979s] Spectrum::CalcFullSpectrum() completed: 5.482 s
==================================================
Total simulation time: 46.981 seconds
```

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```
[14 Spins, 1958M RAM]
[ 0.000s] Program started
[ 0.529s] Hamiltonian constructor completed: 518.8 ms
[ 0.568s] Hamiltonian::Build() completed: 38.2 ms
[ 0.568s] Hamiltonian::FindEigensystem() started...
  Block '1' (size 1x1): 0.0 ms
  Block '2' (size 14x14): 0.0 ms
  Block '3' (size 91x91): 1.4 ms
  Block '4' (size 364x364): 78.1 ms
  Block '5' (size 1001x1001): 2.088 s
  Block '6' (size 2002x2002): 26.165 s
  Block '7' (size 3003x3003): 173.256 s
  Block '8' (size 3432x3432): 157.662 s
  Block '9' (size 3003x3003): 169.296 s
  Block '10' (size 2002x2002): 25.808 s
  Block '11' (size 1001x1001): 2.163 s
  Block '12' (size 364x364): 78.2 ms
  Block '13' (size 91x91): 1.4 ms
  Block '14' (size 14x14): 0.0 ms
  Block '15' (size 1x1): 0.0 ms
[557.165s] Hamiltonian::FindEigensystem() completed: 556.597 s
[649.436s] Hamiltonian::CalcFreqIntens() completed: 92.271 s
Number of Transitions: 204195
Number of Points to calculate: 131072
[665.161s] Spectrum::CalcFullSpectrum() completed: 15.725 s
==============================================
Total simulation time: 665.163 seconds


[15 Spins, 7551M RAM]
[ 0.000s] Program started
[ 2.251s] Hamiltonian constructor completed: 2.238 s
[ 2.387s] Hamiltonian::Build() completed: 135.8 ms
[ 2.387s] Hamiltonian::FindEigensystem() started...
  Block '1' (size 1x1): 0.0 ms
  Block '2' (size 15x15): 0.0 ms
  Block '3' (size 105x105): 2.1 ms
  Block '4' (size 455x455): 201.6 ms
  Block '5' (size 1365x1365): 5.016 s
  Block '6' (size 3003x3003): 162.592 s
  Block '7' (size 5005x5005): 908.313 s
  Block '8' (size 6435x6435): 1784.865 s
  Block '9' (size 6435x6435): 1802.838 s
  Block '10' (size 5005x5005): 938.609 s
  Block '11' (size 3003x3003): 164.202 s
  Block '12' (size 1365x1365): 5.234 s
  Block '13' (size 455x455): 202.7 ms
  Block '14' (size 105x105): 2.0 ms
  Block '15' (size 15x15): 0.0 ms
  Block '16' (size 1x1): 0.0 ms
[5774.465s] Hamiltonian::FindEigensystem() completed: 5772.078 s
[6486.318s] Hamiltonian::CalcFreqIntens() completed: 711.853 s
Number of Transitions: 258114
Number of Points to calculate: 131072
[6506.107s] Spectrum::CalcFullSpectrum() completed: 19.789 s
==============================================
Total simulation time: 6506.109 seconds
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```
[16 Spins, 28.6G RAM]
[ 0.000s] Program started
[ 6.830s] Hamiltonian constructor completed: 6.818 s
[ 7.328s] Hamiltonian::Build() completed: 497.2 ms
[ 7.328s] Hamiltonian::FindEigensystem() started...
  Block '1' (size 1x1): 0.0 ms
  Block '2' (size 16x16): 0.0 ms
  Block '3' (size 120x120): 3.1 ms
  Block '4' (size 560x560): 331.0 ms
  Block '5' (size 1820x1820): 14.857 s
  Block '6' (size 4368x4368): 346.002 s
  Block '7' (size 8008x8008): 2545.940 s
  Block '8' (size 11440x11440): 9486.690 s
  Block '9' (size 12870x12870): 12010.391 s
  Block '10' (size 11440x11440): 9209.779 s
  Block '11' (size 8008x8008): 2459.516 s
  Block '12' (size 4368x4368): 345.406 s
  Block '13' (size 1820x1820): 14.558 s
  Block '14' (size 560x560): 323.1 ms
  Block '15' (size 120x120): 3.0 ms
  Block '16' (size 16x16): 0.0 ms
  Block '17' (size 1x1): 0.0 ms
[36441.128s] Hamiltonian::FindEigensystem() completed: 36433.800 s
[41877.442s] Hamiltonian::CalcFreqIntens() completed: 5436.314 s
Number of Transitions: 554068
Number of Points to calculate: 131072
[41919.909s] Spectrum::CalcFullSpectrum() completed: 42.467 s
==============================================
Total simulation time: 41919.911 seconds
```